Chapter 4. The Fourier Transform

The class $\mathfrak{F}$, Fourier inversion and Poisson summation, the Paley–Wiener theorem

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Overview

For f on $\mathbb{R}$ with suitable regularity and decay, the Fourier transform and its inversion are

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x\xi}\, dx, \qquad f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

The theme of this chapter: the possibility of extending f to a holomorphic function is closely related to very rapid (e.g. exponential) decay of $\hat{f}$ at infinity. First, if f continues analytically to a horizontal strip and has moderate decrease, then $\hat{f}$ decays exponentially, and the inversion and Poisson summation formulas follow by contour integration. Second, assuming only Fourier inversion, the *Paley–Wiener theorem* says $\hat{f}$ is supported in $[-M, M]$ iff f extends to an entire function with $|f(z)| \le Ae^{2\pi M|z|}$ (*exponential type*). The decisive technique is shifting the contour (the real line) within a horizontal strip, exploiting that $e^{-2\pi i z\xi}$ decays or grows exponentially when $\operatorname{Im}(z) \neq 0$, according to the sign of $\xi \operatorname{Im}(z)$.

A function f is of **moderate decrease** if it is continuous and $|f(x)| \le A/(1+x^2)$ for some $A > 0$.

1 The class $\mathfrak{F}$

Book I proved Fourier inversion and Poisson summation when $|f(x)| \le \frac{A}{1+x^2}$ and $|\hat{f}(\xi)| \le \frac{A'}{1+\xi^2}$, a class containing e.g. the Poisson kernel $P_y(x) = \frac{1}{\pi}\frac{y}{y^2+x^2}$, $\hat{P}_y(\xi) = e^{-2\pi y|\xi|}$. Here we introduce a class suited to complex analysis.

Definition (the classes $\mathfrak{F}_a$ and $\mathfrak{F}$)

For $a > 0$, $\mathfrak{F}_a$ is the class of functions f such that:
(i) f is holomorphic in the horizontal strip $S_a = \{z \in \mathbb{C} : |\operatorname{Im}(z)| < a\}$;
(ii) there is $A > 0$ with $|f(x+iy)| \le \dfrac{A}{1+x^2}$ for all $x \in \mathbb{R}$ and $|y| < a$. $\mathfrak{F}$ is the class of functions belonging to $\mathfrak{F}_a$ for some a.

So $\mathfrak{F}_a$ consists of holomorphic functions on S_a of moderate decay on each horizontal line, uniformly in $-a < y < a$. Examples: $e^{-\pi z^2} \in \mathfrak{F}_a$ for all a; $\frac{1}{\pi}\frac{c}{c^2+z^2}$ (simple poles at $\pm ci$) belongs to $\mathfrak{F}_a$ for $0 < a < c$; $1/\cosh \pi z \in \mathfrak{F}_a$ for $|a| < 1/2$. By the Cauchy integral formulas, if $f \in \mathfrak{F}_a$ then $f^{(n)} \in \mathfrak{F}_b$ for $0 < b < a$. (The decrease $A/(1+x^2)$ may be weakened to $A/(1+|x|^{1+\epsilon})$ in most results below.)

2 Action of the Fourier transform on $\mathfrak{F}$

Motivation. For functions that extend holomorphically to a horizontal strip, contour shifting gives clean proofs of the Fourier inversion and Poisson summation formulas, without the convergence difficulties of the real-variable theory.

All three theorems are proved by contour integration.

Theorem 2.1

If f belongs to the class $\mathfrak{F}_a$ for some $a > 0$, then $|\hat{f}(\xi)| \le Be^{-2\pi b|\xi|}$ for any $0 \le b < a$.

Proof. $b = 0$ says $\hat{f}$ is bounded, clear from moderate decrease. Let $0 < b < a$ and $\xi > 0$. Shift the real line down by b: let $g(z) = f(z)e^{-2\pi i z\xi}$ and consider the rectangle with vertices

$-R, R, R-ib, -R-ib$. The integral over the left vertical side is bounded by

$$\Big|\int_{-R-ib}^{-R} g(z)\,dz\Big| \le \int_0^b |f(-R-it)e^{-2\pi i(-R-it)\xi}|\,dt \le \int_0^b \frac{A}{R^2}e^{-2\pi t\xi}\,dt = O(1/R^2),$$

and similarly for the right side. By Cauchy's theorem, letting $R\to\infty$,

$$\hat f(\xi) = \int_{-\infty}^{\infty} f(x-ib)e^{-2\pi i(x-ib)\xi}\,dx, \tag{1}$$

so $|\hat f(\xi)| \le \int_{-\infty}^{\infty}\frac{A}{1+x^2}e^{-2\pi b\xi}\,dx \le Be^{-2\pi b\xi}$. For $\xi<0$ shift the real line up by b. ■

The further f extends (larger a), the better the decay of $\hat f$.

Theorem 2.2

If $f\in\mathfrak{F}$, then the Fourier inversion holds, namely

$$f(x) = \int_{-\infty}^{\infty}\hat f(\xi)e^{2\pi ix\xi}\,d\xi \qquad \text{for all } x\in\mathbb{R}.$$

Lemma 2.3

If A is positive and B is real, then $\int_0^\infty e^{-(A+iB)\xi}\,d\xi = \dfrac{1}{A+iB}$.

Proof. $|e^{-(A+iB)\xi}| = e^{-A\xi}$, so the integral converges, and $\int_0^R e^{-(A+iB)\xi}\,d\xi = \Big[-\frac{e^{-(A+iB)\xi}}{A+iB}\Big]_0^R \to \frac{1}{A+iB}$. ■

Proof of Theorem 2.2. Split $\int_{-\infty}^{\infty}\hat f(\xi)e^{2\pi ix\xi}d\xi = \int_{-\infty}^0 + \int_0^\infty$. Say $f\in\mathfrak{F}_a$, $0<b<a$. By (1) and the lemma (the integrals converge absolutely),

$$\begin{aligned}\int_0^\infty \hat f(\xi)e^{2\pi ix\xi}\,d\xi &= \int_0^\infty\int_{-\infty}^\infty f(u-ib)e^{-2\pi i(u-ib)\xi}e^{2\pi ix\xi}\,du\,d\xi = \int_{-\infty}^\infty f(u-ib)\int_0^\infty e^{-2\pi i(u-ib-x)\xi}\,d\xi\,du\\ &= \int_{-\infty}^\infty f(u-ib)\frac{1}{2\pi b+2\pi i(u-x)}\,du = \frac{1}{2\pi i}\int_{-\infty}^\infty \frac{f(u-ib)}{u-ib-x}\,du = \frac{1}{2\pi i}\int_{L_1}\frac{f(\zeta)}{\zeta-x}\,d\zeta,\end{aligned}$$

$L_1=\{u-ib : u\in\mathbb{R}\}$ traversed left to right. Similarly $\int_{-\infty}^0\hat f(\xi)e^{2\pi ix\xi}d\xi = -\frac{1}{2\pi i}\int_{L_2}\frac{f(\zeta)}{\zeta-x}\,d\zeta$, L_2 the real line shifted up by b, left to right. Now for $x\in\mathbb{R}$ let γ_R be the rectangle with vertices $-R\pm ib$, $R\pm ib$. $f(\zeta)/(\zeta-x)$ has a simple pole at x with residue $f(x)$, so $f(x)=\frac{1}{2\pi i}\int_{\gamma_R}\frac{f(\zeta)}{\zeta-x}\,d\zeta$. As $R\to\infty$ the vertical sides contribute 0, and

$$f(x) = \frac{1}{2\pi i}\int_{L_1}\frac{f(\zeta)}{\zeta-x}\,d\zeta - \frac{1}{2\pi i}\int_{L_2}\frac{f(\zeta)}{\zeta-x}\,d\zeta = \int_0^\infty\hat f(\xi)e^{2\pi ix\xi}\,d\xi + \int_{-\infty}^0\hat f(\xi)e^{2\pi ix\xi}\,d\xi.$$

■

Theorem 2.4 (Poisson summation formula)

If $f\in\mathfrak{F}$, then $\displaystyle\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}}\hat f(n)$.

Proof. Say $f\in\mathfrak{F}_a$, $0<b<a$. $1/(e^{2\pi iz}-1)$ has simple poles with residue $1/(2\pi i)$ at the integers, so $f(z)/(e^{2\pi iz}-1)$ has simple poles at the integers n with residues $f(n)/2\pi i$. Apply the residue formula to the rectangle γ_N with vertices $\pm(N+\frac12)\pm ib$:

$$\sum_{|n|\le N} f(n) = \int_{\gamma_N}\frac{f(z)}{e^{2\pi iz}-1}\,dz.$$

Let $N \to \infty$; by moderate decrease the vertical sides contribute 0, and

$$\sum_{n\in\mathbb{Z}} f(n) = \int_{L_1} \frac{f(z)}{e^{2\pi i z}-1}\,dz - \int_{L_2} \frac{f(z)}{e^{2\pi i z}-1}\,dz, \tag{2}$$

L_1, L_2 the real line shifted down and up by b. If $|w| > 1$, $\frac{1}{w-1} = w^{-1}\sum_{n=0}^{\infty} w^{-n}$, so on L_1 (where $|e^{2\pi i z}| > 1$) $\frac{1}{e^{2\pi i z}-1} = e^{-2\pi i z}\sum_{n=0}^{\infty} e^{-2\pi i n z}$; if $|w| < 1$, $\frac{1}{w-1} = -\sum_{n=0}^{\infty} w^n$, so on L_2 $\frac{1}{e^{2\pi i z}-1} = -\sum_{n=0}^{\infty} e^{2\pi i n z}$. Substituting in (2) and shifting L_1, L_2 back to the real line (by (1) and its analogue),

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{n=0}^{\infty}\int_{L_1} f(z)e^{-2\pi i(n+1)z}\,dz + \sum_{n=0}^{\infty}\int_{L_2} f(z)e^{2\pi i n z}\,dz = \sum_{n=0}^{\infty}\hat{f}(n+1) + \sum_{n=0}^{\infty}\hat{f}(-n) = \sum_{n\in\mathbb{Z}}\hat{f}(n).$$

■

Consequences. Since $e^{-\pi x^2}$ is its own Fourier transform (Example 1), for $t > 0$ and $a \in \mathbb{R}$ the change of variables $x \mapsto t^{1/2}(x+a)$ shows that $f(x) = e^{-\pi t(x+a)^2}$ has $\hat{f}(\xi) = t^{-1/2}e^{-\pi\xi^2/t}e^{2\pi i a\xi}$. Poisson summation gives

$$\sum_{n=-\infty}^{\infty} e^{-\pi t(n+a)^2} = \sum_{n=-\infty}^{\infty} t^{-1/2}e^{-\pi n^2/t}e^{2\pi i n a}. \tag{3}$$

For $a = 0$ this is the transformation law of the theta function $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$ $(t > 0)$:

$$\vartheta(t) = t^{-1/2}\vartheta(1/t) \qquad \text{for } t > 0, \tag{4}$$

used in Chapter 6 for the functional equation of ζ; the general case is used in Chapter 10 for the Jacobi theta function. Likewise, since $1/\cosh \pi x$ is its own Fourier transform (Example 3), $f(x) = e^{-2\pi i a x}/\cosh(\pi x/t)$ has $\hat{f}(\xi) = t/\cosh(\pi(\xi+a)t)$, and

$$\sum_{n=-\infty}^{\infty} \frac{e^{-2\pi i a n}}{\cosh(\pi n/t)} = \sum_{n=-\infty}^{\infty} \frac{t}{\cosh(\pi(n+a)t)}, \tag{5}$$

used in Chapter 10 for the two-squares theorem.

3 Paley–Wiener theorem

Motivation. Which functions have Fourier transforms supported in a bounded interval? Exactly those that extend to entire functions of exponential type. The theorem shows how decay on one side of the Fourier transform corresponds to analyticity on the other.

Now we assume no analyticity, but the validity of Fourier inversion when $|f(x)| \le A/(1+x^2)$ and $|\hat{f}(\xi)| \le A'/(1+\xi^2)$ (proved in Book I). A partial converse to Theorem 2.1:

Theorem 3.1

Suppose $\hat{f}$ satisfies the decay condition $|\hat{f}(\xi)| \le Ae^{-2\pi a|\xi|}$ for some constants $a, A > 0$. Then $f(x)$ is the restriction to $\mathbb{R}$ of a function $f(z)$ holomorphic in the strip $S_b = \{z \in \mathbb{C} : |\mathrm{Im}(z)| < b\}$, for any $0 < b < a$.

Proof. Let $f_n(z) = \int_{-n}^{n} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$, entire by Theorem 5.4. For $z \in S_b$ define $f(z) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$; the integral converges absolutely, being majorized by $A\int e^{-2\pi a|\xi|}e^{2\pi b|\xi|}\,d\xi < \infty$ as $b < a$. Moreover for $z \in S_b$, $|f(z) - f_n(z)| \le A\int_{|\xi|\ge n} e^{-2\pi a|\xi|}e^{2\pi b|\xi|}\,d\xi \to 0$, so $f_n \to f$ uniformly in S_b, and f is holomorphic there by Theorem 5.2. ■

Corollary 3.2

If $\hat{f}(\xi) = O(e^{-2\pi a|\xi|})$ for some $a > 0$, and f vanishes in a non-empty open interval, then $f = 0$.

f is analytic in a region containing $\mathbb{R}$, so this follows from Theorem 4.8. In particular f and $\hat{f}$ cannot both have compact support unless $f = 0$.

Theorem 3.3 (Paley–Wiener)

Suppose f is continuous and of moderate decrease on $\mathbb{R}$. Then, f has an extension to the complex plane that is entire with $|f(z)| \leq Ae^{2\pi M|z|}$ for some $A > 0$, if and only if $\hat{f}$ is supported in the interval $[-M, M]$.

Proof. *($\Leftarrow$)* If $\hat{f}$ is supported in $[-M, M]$, both f and $\hat{f}$ have moderate decrease and $f(x) = \int_{-M}^{M} \hat{f}(\xi)e^{2\pi i\xi x}\,d\xi$. Define $g(z) = \int_{-M}^{M} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$; $g = f$ on $\mathbb{R}$, g is entire by Theorem 5.4, and for $z = x + iy$, $|g(z)| \leq \int_{-M}^{M} |\hat{f}(\xi)|e^{-2\pi\xi y}\,d\xi \leq Ae^{2\pi M|z|}$.

($\Rightarrow$) If $\hat{f}$ were supported in $[-M, M]$, the argument above would give the stronger bound $Ae^{2\pi M|y|}$. We reduce to that situation, adding also decay in x, and then remove the extra assumptions step by step.

Step 1. Assume f is entire with

$$|f(x + iy)| \leq A\frac{e^{2\pi M|y|}}{1 + x^2}. \tag{6}$$

Then $\hat{f}(\xi) = 0$ for $|\xi| > M$. For $\xi > M$, shifting the real line down by $y > 0$ (as in (1)),

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x - iy)e^{-2\pi i(x-iy)\xi}\,dx, \qquad |\hat{f}(\xi)| \leq \int_{-\infty}^{\infty} A\frac{e^{2\pi My - 2\pi y\xi}}{1 + x^2}\,dx \leq Ce^{-2\pi y(\xi - M)},$$

and letting $y \to \infty$ gives $\hat{f}(\xi) = 0$. For $\xi < -M$ shift up.

Step 2. Assume only

$$|f(x + iy)| \leq Ae^{2\pi M|y|}. \tag{7}$$

Let $\xi > M$ and for $\epsilon > 0$ set $f_\epsilon(z) = \dfrac{f(z)}{(1 + i\epsilon z)^2}$. The factor $1/(1 + i\epsilon z)^2$ has absolute value ≤ 1 in the closed lower half-plane and tends to 1 as $\epsilon \to 0$; hence $\hat{f}_\epsilon(\xi) \to \hat{f}(\xi)$, since $|\hat{f}_\epsilon(\xi) - \hat{f}(\xi)| \leq \int |f(x)|\left|\frac{1}{(1+i\epsilon x)^2} - 1\right| dx \to 0$ by moderate decrease. Each f_ϵ satisfies $|f_\epsilon(x + iy)| \leq A'\frac{e^{2\pi M|y|}}{1+x^2}$ in the lower half-plane, which is where Step 1 shifts the contour for $\xi > M$; so $\hat{f}_\epsilon(\xi) = 0$, and hence $\hat{f}(\xi) = 0$. For $\xi < -M$ argue in the upper half-plane with $1/(1 - i\epsilon z)^2$.

Step 3. It remains to show the hypotheses imply (7). Dividing by a constant, it suffices: if $|f(x)| \leq 1$ for real x and $|f(z)| \leq e^{2\pi M|z|}$ for all z, then $|f(x+iy)| \leq e^{2\pi M|y|}$. This follows from the Phragmén–Lindelöf theorem below, rotated to the first quadrant $Q = \{x > 0, y > 0\}$: $F(z) = f(z)e^{2\pi iMz}$ is bounded by 1 on the positive real and imaginary axes and $|F(z)| \leq Ce^{c|z|}$ in Q, so $|F(z)| \leq 1$ in Q, i.e. $|f(z)| \leq e^{2\pi My}$. The other quadrants are similar. ∎

The idea of Phragmén and Lindelöf adapts the maximum modulus principle to unbounded regions:

Theorem 3.4 (Phragmén–Lindelöf)

Suppose F is a holomorphic function in the sector $S = \{z : -\pi/4 < \arg z < \pi/4\}$ that is continuous on the closure of S. Assume $|F(z)| \leq 1$ on the boundary of the sector, and that there are constants $C, c > 0$ such that $|F(z)| \leq Ce^{c|z|}$ for all z in the sector. Then $|F(z)| \leq 1$ for all $z \in S$.

Some growth restriction is needed: $F(z) = e^{z^2}$ is bounded by 1 on ∂S but unbounded on $\mathbb{R}_+$.

Proof. Subdue the "enemy" e^{z^2}: replace it by e^{z^α} with $\alpha < 2$; take $\alpha = 3/2$. For $\epsilon > 0$ let $F_\epsilon(z) = F(z)e^{-\epsilon z^{3/2}}$, with the principal branch, so $z^{3/2} = r^{3/2}e^{3i\theta/2}$ for $z = re^{i\theta}$, $-\pi < \theta < \pi$. F_ϵ is holomorphic in S and continuous up to the boundary, and $|e^{-\epsilon z^{3/2}}| = e^{-\epsilon r^{3/2}\cos(3\theta/2)}$. In the sector $-\pi/4 < \theta < \pi/4$ we have $-\frac{\pi}{2} < -\frac{3\pi}{8} < \frac{3\theta}{2} < \frac{3\pi}{8} < \frac{\pi}{2}$, so $\cos(3\theta/2)$ is bounded below by a positive constant. With $|F(z)| \le Ce^{c|z|}$, F_ϵ decreases rapidly as $|z| \to \infty$ in the closed sector and is bounded. Let $M = \sup_{z\in\overline{S}}|F_\epsilon(z)|$; if $F \not\equiv 0$, take w_j with $|F_\epsilon(w_j)| \to M$. Since $M \neq 0$ and F_ϵ decays at infinity, the w_j accumulate at some $w \in \overline{S}$. By the maximum principle $w \notin S$, so $w \in \partial S$, where $|F| \le 1$ and $|e^{-\epsilon z^{3/2}}| \le 1$; so $M \le 1$. Thus $|F_\epsilon| \le 1$ on $\overline{S}$, and letting $\epsilon \to 0$ gives $|F| \le 1$. ■

Another version of the Paley–Wiener idea characterizes Fourier transforms vanishing for $\xi < 0$:

Theorem 3.5

Suppose f and $\hat{f}$ have moderate decrease. Then $\hat{f}(\xi) = 0$ for all $\xi < 0$ if and only if f can be extended to a continuous and bounded function in the closed upper half-plane $\{z = x + iy : y \ge 0\}$ with f holomorphic in the interior.

Proof. If $\hat{f}(\xi) = 0$ for $\xi < 0$, then $f(x) = \int_0^\infty \hat{f}(\xi)e^{2\pi i\xi x}\,d\xi$, and we extend f for $y \ge 0$ by $f(z) = \int_0^\infty \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$. The integral converges, and $|f(z)| \le A\int_0^\infty \frac{d\xi}{1+\xi^2} < \infty$, so f is bounded. The functions $f_n(z) = \int_0^n \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$ converge to f uniformly in the closed half-plane, which gives continuity there and holomorphy in the interior.

Conversely, in the spirit of Theorem 3.3, for $\epsilon, \delta > 0$ set $f_{\epsilon,\delta}(z) = \dfrac{f(z + i\delta)}{(1 - i\epsilon z)^2}$, holomorphic in a region containing the closed upper half-plane. As before, by Cauchy's theorem (shifting the contour up), $\hat{f}_{\epsilon,\delta}(\xi) = 0$ for all $\xi < 0$. Passing to the limit successively, $\hat{f}_{\epsilon,0}(\xi) = 0$ for $\xi < 0$, and finally $\hat{f}(\xi) = \hat{f}_{0,0}(\xi) = 0$ for $\xi < 0$. ■

Remark

Compare Theorem 7.1: there, for a function holomorphic in a disc, the Fourier coefficients vanish for $n < 0$; here, for a function holomorphic in the upper half-plane, the Fourier transform vanishes for $\xi < 0$.